Workshop warm-up tasks

Applications of Machine Learning Workshop

EPFL Extension School

Our workshop has three sections. Below you find a warm up task for each of the sections. Please read each task and note your observations. They will help prepare your thinking for some of the central questions in the workshop.

1 Text analysis

Part 1

The following words are a "summary" of the most relevant part of the email.

```
october subject machine_learning workshop attend workshop workshop catch offer epfl
website familiar machine_learning
```

> Questions
>
> - **Question 1**: What are the most frequent words in the summary?
> - **Question 2**: Would you be able to guess the topic of the email?

Part 2

Here is the original email text, but formatted at the very basic level of the email client:

```
--------------- >From: nick@gmail.com [mailto: nick@gmail.com] >Sent: 16 October 2022
17:42 ---------------->To: 'james@gmail.com >Subject: machine learning workshop
---------------> Hi James, how are you doing? </>Are you attending the workshop? <>
The title of the workshop is catchy. It is offered by EPFL, here is their website
 https://www.extensionschool.ch> By the way, are you familiar with machine learning?...
```

Questions

- **Question 3**: What are the most relevant parts of this text to extract the meaning of the email?
- **Question 4**: What are the most relevant parts of this text to decide whether the email is spam or genuine?

2 Anomaly detection

On your way to/from work passively observe the people and the traffic around you.

Questions

- **Question 5 v1**: If you are taking public transport: What made a person stand out for you?
- **Question 5 v2**: If you are driving: What made a car stand out for you?

3 Large Language Models

Attached are two chat protocols between Adam and the customer service of an online shop selling tires.

Question 6

Read the original protocol (`Chatbot - original conversation.pdf`) and collect your observations. What are positive points or points for improvement?

LLMs can translate text for us, so we passed the text through an LLM to translate all French parts into English.

Question 7

Read the translated version (`Chatbot - ENG translation.pdf`) and collect your observations. Again what are positive points or points for improvement?